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MILITARY SPECIFICATION

WIRE AND CABLE, POLYIMIDE INSULATED, ELECTRICAL NUCLEAR WEAPONS

1. SCOPE

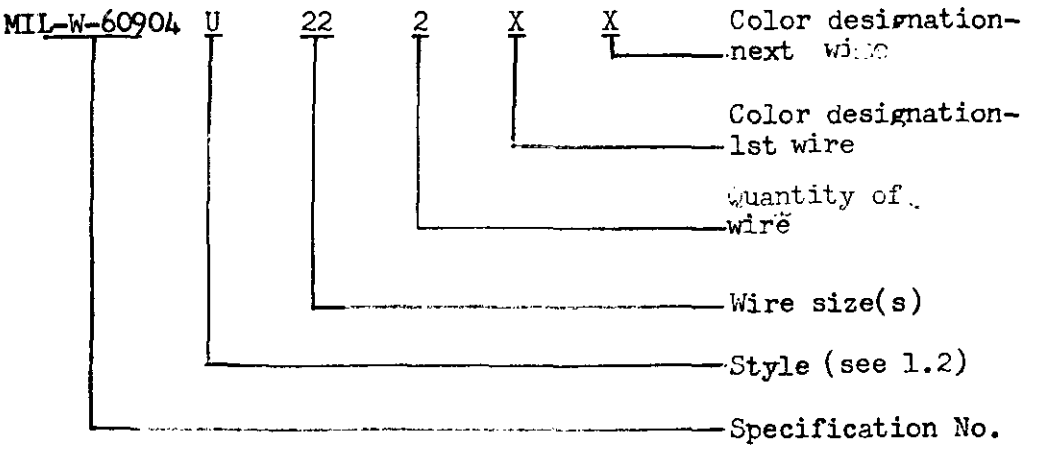
1.1 Scope. - This specification covers special purpose polyimide-insulated and shielded wire and cable for hook-up and inter-connections.

1.2 Classification. - Completed wire and cable in accordance with this specification shall be of styles as follows:

"U" - Unshielded
"S" - Shielded and unjacketed
"SJ" - Shielded and jacketed

The conductor sizes shall be American Wire Gauge (AWG) 26 through 16 specified in Table I (see 6.2).

1.3 Identification. - The identification of completed wire and cable shall be a combination of digits and letters as specified.



FSC-1135

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2. APPLICABLE DOCUMENTS

2.1 *Documents.* - The following documents in effect on date of invitation for bids, form a part of this purchase description to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-W-343 - Wire, Electrical and Nonelectrical, Copper Uninsulated

MILITARY

MIL-D-26937 - Detergent, Synthetic, Anionic Alkyl Benzene Sulfonate
MIL-P-46112 - Plastic Sheet and Strip, Polyimide.

STANDARDS

MILITARY

MIL-STD-105 - Sampling procedures and tables for Inspection by Attributes.
MIL-STD-109 - Quality Assurance terms and definitions.
MIL-STD-681 - Identification, Coding and Application of Hook-up Wire.
MIL-STD-1169 - Packaging, Packing and Marking for shipments of Inert Ammunition Components.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

NON-GOVERNMENT

ASTM B-298 - Standard Specification for Silver Coated Soft Annealed Copper Wire.

(ASTM - American Society for Testing and Materials. ASTM specifications may be obtained by addressing application to: ASTM, 1916 Race Street, Philadelphia, Pennsylvania.)

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3. REQUIREMENTS

3.1 Material and assembled wire and cable. - All materials used shall comply with applicable specifications and completed wire and cable with this specification. The contractor shall have objective evidence (see 6.2) available, proving that materials used comply, that processes, fabrications and inspections were accomplished under controllable conditions as set forth in the contractor's quality assurance procedures.

3.1.1 Conductor and Shield. - All strands used in the manufacture of the conductor shall be silver coated soft annealed copper conforming to ASTM-B-298, except that coating thickness shall be not less than 40 micro inches. Strands shall be free from lumps, kinks, splits, abrasions and coating impurities. All strands used in the manufacture of the shield of style "S" and "SJ" wire and cable shall conform to QQ-W-343, Type S.

3.1.2 Insulation. - Insulation shall be formed from Polyimide Tape, MIL-P-46112, Type II, Grade A, except that outer wrap of style "SJ" cable shall be Type II, Grade B.

3.1.2.1 Color. - Color, when specified, shall be stripes, conforming to MIL-STD-681, as indelibly applied on the FEP Fluoro Carbon - coated tape surface of the underside of the outer layer of Polyimide tape. The color, when viewed on the finished wire or cable, will not necessarily conform to MIL-STD-681.

3.2 Construction Properties

3.2.1 Conductor

3.2.1.1 Stranding. - Stranding shall be unidirectional concentric lay (Unilay) which is defined as a central core surrounded by one or more concentric layers of helically wound strands. Each successive layer of strands shall be in the same direction. The strands shall be assembled so as to provide a smooth uniform assembly with a circular cross section and free from surface irregularities. The outer layer shall have a lay length not less than 8 or more than 16 times the maximum diameter of the conductor. Stranded conductor sizes shall be as specified in Table I.

Table I

Number of Strands X Strand Size (AWG)	Stranded Conductor Size (AWG)	Conductor Diameter Inch, MAX.
19X38	26	0.021
19X36	24	0.026
19X34	22	0.033
19X32	20	0.041
19X30	18	0.052
19X29	16	0.060

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3.2.1.1.1 Strand splices - Splices in individual strands shall be brazed. There shall be not more than one splice in any 10 foot length of stranded conductor. In no case shall all the strands be spliced at one point.

3.2.1.2 Insulation wrap

3.2.1.2.1 Insulation - style "U" and style "SJ" wire. - The first layer of tape, with the FEP Fluorocarbon-coated surface facing away from the conductor, shall be applied to assure a 50 percent min overlap. The second layer of tape, with the FEP Fluorocarbon facing toward the conductor shall be applied with a reverse wrap in relation to the first layer to assure a 30 percent min overlap.

3.2.1.2.2 Insulation - style "SJ" cable. - The first layer of tape, with the FEP Fluorocarbon-coated surface facing away from the wires, shall be applied to assure a 50 percent min overlap. The second layer of tape (see 3.1.2) shall be reverse wrapped in relation to the first layer to assure a 30 percent min overlap.

3.2.1.2.3 Thickness. - The insulation along the length of the insulated conductor at any point, shall be equal to three thickness min of tape.

3.2.1.2.4 Splices. - There shall be not more than one splice in the outer insulation for every 50 feet of insulated conductor.

3.2.2 Shield. - The wire shall be applied at an angle of 25±15 degrees ($^{\circ}$) to provide an 85 percent coverage min as determined by the following formula.

Formula

$$K = 100\ (2F - F^2)$$

where:

K = Percent coverage
F = N Pd/Sin a
N = Number of Strands per carrier
P = Picks per inch of wire or cable length
d = Diameter of one of the shield strands in inches
a = Angle of Board with axis of wire or cable

$$\text{Tan } a = 2\pi (D + 2d)\ P/C$$

C = Number of carriers
D = Diameter of wire or cable under braid in inches, except that for 2 conductor cable only,

$$D = \frac{(\pi + 2)}{\pi}$$

where: b = diameter of one wire

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The shield strand size is based on the size of the wire and shall be as specified in Table II.

Table II

Wire Size (AWG)	Strand Size (AWG)
26 through 22	38
20 and larger	36

3.2.3 Jacket (see 3.2.1.2.2).

3.2.4 Wire and Cable Diameter (see 1.2 and 6.2). - Wire diameters shall be as specified in Tables III, IV, and V.

Table III

Style "U" - Conductor Size (AWG)	Wire Outside Diameter Inches - MAX.
26	0.032
24	0.036
22	0.043
20	0.051
18	0.060
16	0.067

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Table IV

Quantity of Style "U" Wire	Style "S" Wire and Cable Outside Diameter - Inches, MAX.					
1	0.051	.055	.062	.074	.083	0.090
2	0.083	.091	.105	.125	.143	0.157
3	0.088	.097	.112	.133	.153	0.168
4	0.097	.106	.123	.146	.168	0.185
7	0.115	.127	.148			
Style "U" Wire Size (AWG)	26	24	22	20	18	16

Table V

Quantity of Style "U" Wires	Style "SJ" Wire and Cable Outside Diameter - Inches, MAX.					
1	0.065	.069	.076	.088	.097	0.104
2	0.097	.105	.119	.139	.157	0.171
3	0.102	.111	.126	.147	.167	0.182
4	0.111	.120	.137	.160	.182	0.199
7	0.129	.141	.162			
Style "U" Wire Size (AWG)	26	24	22	20	18	16

3.2.4.4 Lay Length. - A cable having 2 or more wires shall have a lay length not less than 8 or more than 14 times the overall diameter of the cabled wire.

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3.3 Electrical and Mechanical Characteristics. - Completed wire and cable shall conform to the following requirements.

3.3.1 Dielectric strength (dry). - All completed wire (Style "U") and cable (Style "SJ") shall withstand a potential of 2000 volts, alternating current (vac) min at 60 cycles per second (cps) impressed for one minute min without breakdown. Testing shall be as specified in 4.4.01.

3.3.2 Jacket flaw. - All style "SJ" wire and cable shall withstand a potential of 1500 vac min at 60 cps impressed for 0.2 second min without breakdown. Testing shall be as specified in 4.4.02.

3.3.3 Heat resistance. - All completed wire and cable shall withstand a temperature of +482±10 degrees Fahrenheit (°F) for 6 hours, min, followed by a bend and dielectric test without breakdown or cracking of materials. Testing shall be as specified in 4.4.03.

3.3.4 Dielectric strength (tank). - All completed wire (Style "U") and cable (Style "SJ") shall withstand a potential of 2000 vac min at 60 cps impressed for one minute min without breakdown. Testing shall be as specified in 4.4.04.

3.3.5 Cold bend. - All completed wire (Style "U") and cable (Style "SJ") shall withstand bend stresses while the temperature is maintained at -65±2°F and dielectric strength without insulation breakdown or cracking of material. Testing shall be as specified in 4.4.05.

3.3.6 Continuity. - Conductors forming completed wire and cable shall indicate electrical continuity. Testing shall be as specified in 4.4.06.

3.3.7 Resistance. - Conductors forming completed wire and cable shall indicate electrical resistance as described in Table VI and VIA while the temperature is maintained at +68±5°F. Testing shall be as specified in 4.4.07.

Table VI

Size (AWG) of Only a Single Wire Style "U" or Style "SJ"	Ohms max per 1000 feet
26	37.30
24	23.60
22	14.80
20	9.09
18	5.80
16	4.54

Table VI-A

Size (AWG) of each wire in completed cable	Ohms max per 1000 feet
26	38.42
24	24.31
22	15.24
20	9.36
18	5.97
16	4.68

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3.3.8 Solder resistance. - The insulating material shall not shrink back along the conductor more than 1/16 inch, flare away or open up over the bend where the end of the sample is immersed in 60-40 solder maintained at temperatures between +590°F and +608°F. Testing shall be as specified in 4.4.08.

3.3.9 Delamination. - All completed wire and cable shall withstand a temperature of +482 ± 10°F. for 6 ± 1/2 hour min without separation of layers either along the insulation or at both ends of the sample. Testing shall be as specified in 4.4.10.

3.3.10 Workmanship. - All completed wire and cable shall be finished and packaged in a thoroughly workmanlike manner. Finished cable shall be free from lumps, kinks and the insulation, shield and jacket shall not be broken exposing the underneath material.

3.3.11 Lengths. - All completed wire and cables shall be shipped in lengths not less than 100 nor more than 1000 feet (see 5.1.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification when such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Inspection terms are defined in MIL-STD-109.

4.2 First article approval. - Not applicable.

4.3 Inspection provisions.

4.3.1 Inspection lot. - Lot or batch and its formation, size, length and presentation are described in MIL-STD-105, section 5 for the purpose of this specification, "Lot" shall mean inspection lot and "Batch" shall mean inspection batch. The manner in which each inspection lot or batch is to be presented and identified by the contractor shall be designated or approved by the Government representative.

4.3.2 Inspection lot approval. - The inspections prescribed under 4.3.2.1 and 4.3.3 shall be accomplished with Government approved equipment (see 4.3.4).

4.3.2.1 Classification of defects. - Inspection shall be performed 100 percent for all defect classification unless specified otherwise. One defect shall reject the entire length of completed wire and cable.

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4.3.2.1.1 Conductor (see 3.1.1)

Categories	Defects	Method of Inspection
Critical: None defined.		
Major:		
	101. Objective evidence (see 6.2) that material conforms to QQW-343 is missing or improper	Visual
	102. Incorrect thickness of silver coating (see 3.1.1)	Gage
Minor: None defined.		

4.3.2.1.2 Finished wire and cable.

Categories	Defects	Method of Inspection
Critical: None defined.		
Major:		
	101. Color code missing, improper or unidentifiable (see 3.1.2) as viewed through the outer layer when applicable	Visual
	*102. Incorrect overlap of insulation (see 3.2.1.2) or jacket (see 3.2.3) as appropriate	Gage
	*103. Incorrect diameter of conductor (see 3.2.1.1)	Gage
	104. Conductor not free from high stranding or surface irregularity (see 3.2.1.1)	Visual
	105. Excessive or improper splicing (see 3.2.1.1.1) and 3.2.1.2.4)	Visual
	106. Improper insulation (see 3.1.2 as appropriate)	Visual
	*107. Shield improperly applied (see 3.2.2)	Visual
	108. Incorrect outside diameter of style "U" (see 3.2.4.) style "S" (see 3.2.4.) or style "SJ" (see 3.2.4. as appropriate)	Gage
	109. Evidence of poor workmanship (see 3.3.10)	Visual

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Minor: None defined.

*Verify or perform this inspection on the "first piece" sample which the contractor shall take at the start and at the end of each production shift.

4.3.2.1.3 Preparation for delivery (see 5.0).

Categories	Defects	Method of Inspection

Critical: None defined.

Major:

101.	Incorrect total length of completed wire or cable on spool or reel - Before specimen has been cut-off (see 4.4.09)	Gage
102.	Incorrect total length of completed wire or cable on spool or reel - After specimen has been cut-off (see 3.11)	Gage
103.	Improper type of spool or reel (see 5.1.1)	Gage
104.	Improper container (see 5.2)	Visual
105.	Improper finish (see 5.1.1)	Visual
106.	Damaged finish exposing base metal	Visual
107.	Flange not properly attached to barrel (see 5.1.1)	Visual
108.	Damaged spool or reel (only if damage can inflect injury to personnel)	Visual
109.	Marking missing, improper, illegible or misplaced (see 5.3)	Visual

Minor: None defined.

4.3.3 Inspection testing. - The contractor shall perform all the tests specified in this paragraph. The Government inspector will, as a minimum, witness the actual performance of the testing conducted by the contractor; independently determine conformance to the prescribed requirements and independently record the results obtained: verify that the contractor's performance in the preparation of samples for preconditioning and placement in preconditioning equipment is accomplished in accordance with the prescribed requirement; and monitor the activity and equipment throughout the prescribed conditioning cycle to insure conformance in that regard. The Government inspector will not be limited to witnessing the contractor's performance of the test, but may, at his own discretion, perform any of the tests independently. All wire and cable shall be subjected to the tests specified in the Table VII.

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TABLE VII

TESTS (See Note 3)	Quantity Wire & Cable See:	Requirement See:	Testing Method See:	Finished Wire and Cable		
				Before Shielding or Jacketing	After Shielding	After Shielding and Jacketing
(a)Dielectric (dry)	Note 1	3.3.1	4.4.01			X
(b)Jacket flaws	Note 1	3.3.2	4.4.02	X		X
(c)Continuity	Note 1	3.3.6	4.4.06	X	X	X
(d)Resistance	Note 1	3.3.7	4.4.07	X	X	X
(e)Dielectric (tank)	Note 1	3.3.4	4.4.04	X		X
(f)Delamination	Note 2	3.3.9	4.4.10	X		X
(g)Solder resistance	Note 2	3.3.8	4.4.08	X		X
(h)Heat resistance	Note 2	3.3.3	4.4.03	X		X
(i)Cold bend	Note 2	3.3.5	4.4.05	X		X

Note 1 - See page 12
Note 2 - See page 12
Note 3 - Tests (f) through (i) shall be conducted in the order tabulated.

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NOTE 1. The finished wire or cable shall be subjected to this test. If the wire or cable fails to comply with any requirements when testing as specified in the applicable paragraph under 4.4.00, the entire length of finished wire or cable shall be rejected.

NOTE 2. Two samples shall be taken as described under 4.4.00 and both shall be subjected to this test. If either sample fails to comply with any requirement when testing as specified in the applicable paragraph under 4.4.00, the entire length of finished cable or wire shall be rejected.

4.3.4 Inspection equipment. -

4.3.4.1 Equipment. - Inspection and testing shall be accomplished with the equipment and instructions approved by the procuring activity.

4.4.00 Test methods and procedures. - The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used. All tests will be conducted at ambient temperature (80±20°F) unless otherwise specified.

4.4.01 Dielectric strength (dry). - One hundred percent of completed style "U" wire and style "CU" cable shall be subjected to 2000 vac min at 60 cps impressed at a uniform rate so that 2000 vac is reached within 1/2 minute, maintain for 1 minute, then reduced to zero volts within 1/2 minute at a uniform rate. A suitable testing device with its current limited to 100 milliamperes (ma) shall be used to determine compliance with 3.3.1. In Style "CU" cable, select one wire, as a terminal, and connect all other wires and shield together to form the other terminal. Apply test potential, as described above, and repeat test in turn with each successive wire.

4.4.02 Jacket flaw (Spark test). - One hundred per cent of all completed wire and cable, except style "U" shall be subjected to 1500 vac min at 60 cps impressed for 0.2 second min. A suitable testing device, of the bead-chain electrode type with its current limited to 100 ma max. shall be used to determine compliance with 3.3.2. The electrode shall be constructed to assure intimate metallic contact with practically all the wire insulated surface. Electrode length and speed of wire movement shall be such that the insulation is subjected to the test voltage and for the time duration specified.

4.4.03 Heat resistance. - A 12 inch sample shall be cut from each specimen length (see 4.4.09). One inch shall be stripped from each end of the sample and bent at the middle of the length 180° min around a mandrel. Apply a weight at each end of the bared conductor in such a manner that the portion of the outer insulation between the cable and mandrel is under compression while the conductor is under tension. The combination of mandrel size and test load shall be as specified in table VIII. Each prepared sample then shall be placed into an air circulating temperature chamber. The temperature shall be raised to 482±10°F and maintained for six hours, min. While being conditioned, air shall be induced and regulated so that it passes around the sample at a velocity rate between 100 and 200 feet per minute. After removal from the chamber, the sample shall be cooled to 473±5°F within one hour. After cooling, release the tension, remove the mandrel and straighten

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the cable. The entire surface of the outer jacket shall be visually inspected to determine compliance with 3.3.3. The sample then, shall be tested for compliance with bend 4.4.03.1 and wet dielectric 4.4.03.2.

4.4.03.1 Bend. - While the sample is maintained at +73±5°F, one end shall be secured to the mandrel and the other end to the load weight. The mandrel shall be rotated until the full length of the sample is wrapped around the mandrel with adjoining coils in contact. The mandrel now shall be rotated in reverse direction until the full length of the cable which was outside during the first wrapping is now next to the mandrel. This procedure shall be repeated until two bends in each direction have been formed in the same direction of the cable. The cable shall be straightened and visually inspected to determine compliance with 3.3.3.

4.4.03.2 Wet dielectric. - The uninsulated ends of the sample shall be fastened in metallic contact to a metal bar. That area of the sample which was wound around the mandrel shall be bent in the shape of a "U", the diameter of the bend to be not less than three times the diameter of the cable. The sample then, shall be immersed in a five-percent solution of sodium chloride in water at temperatures ranging between +68 and +77°F so that the jacketed portion of the cable protrudes three inches from the surface of the liquid. A suitable testing device with current limited to 100 ma max shall be used to determine compliance with 3.3.3.

4.4.03.2.1 Style "U". - Each sample shall be immersed for five hours. Then subjected to 2000 vac min at 60 cps impressed at a uniform rate so that 2000 vac is reached within 1/2 minute. Maintain for five minutes min, reduce to zero volt within 1/2 minute at a uniform rate.

4.4.03.2.2 Style "SJ". - Each sample shall be immersed for one hour. Then subject to 1000 vac min at 60 cps impressed at a uniform rate so that 1000 vac is reached within 1/2 minute. Maintain for one minute min, then reduce to zero volts within 1/2 minute at a uniform rate.

4.4.04 Dielectric strength (tank). - One hundred percent of all completed wire and cable (except style "S") spooled or reeled shall be immersed in a bath containing 0.5 to 1.0 percent by volume of anionic wetting agent for 4 hours. The anionic wetting shall conform to MIL-D-26937. During this test, both ends of each length of completed wire or cable under test shall be connected to the high voltage terminal and the solution shall be grounded and connected to the other end of the high voltage transformer winding. After the wire or cable has been immersed for four hours min an AC potential of 2000 volts min shall be applied. The AC potential shall be increased gradually from zero at a rate not exceeding 500 volts per second. After being held at the full potential for 1 minute min, the voltage shall be decreased gradually to zero at the same rate. A suitable testing device with its current limited to 100 ma max shall be used to determine compliance with 3.3.4. It is permissable to wind completed wire or cable on testing spools or reels for this test and to rewind on shipping spools or reels after completion of this test.

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TABLE VIII

MANDREL DIAMETERS AND WEIGHTS

One Conductor Size (AWG)	Mandrel Diameter (inches-Max.)				Test Load (lbs.)			
	Heat Resistance and Bend		Cold Bend		Heat Resistance and Bend		Cold Bend	
	Type U	Type SJ	Type U	Type SJ	Type U	Type SJ	Type U	Type SJ
26	1/8	5/16	1/4	5/8	1/2	3/4	3/4	1
24	3/16	3/8	3/8	3/4	3/4	3/4	3/4	1
22	3/16	3/8	3/8	3/4	3/4	1	1	2
20	1/4	7/16	1/2	7/8	3/4	1	1	2
18	1/4	7/16	1/2	7/8	2	2	2	3
16	5/16	1/2	5/8	1	2	2	2	3
More than 1 Conductor	Six times cable diameter							

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4.4.05 Cold bend. - A 24 inch sample shall be cut from each specimen length (see 4.4.09). Each 24 inch sample shall be prepared so that one end is secured to a rotatable mandrel and the other end to a weight. The combination of mandrel size and weight load shall be selected as prescribed in table VIII. Proper arrangement shall be made for rotating the mandrel by means of a handle or control located outside the chamber. Each 24 inch samples and its mandrel then shall be placed into an approved temperature chamber and conditioned at -67±2°F for 4 hours and until the temperature of the mandrel has stabilized (see 6.3). (The temperature shall be lowered to -67±2°F at a rate not exceeding 90°F per minute). After stabilization, each 24 inch sample shall be wrapped around its mandrel 180° degrees min for style "U" and 720° min for Style "SJ". Wrapping around the mandrel shall be accomplished within 1/2 minute at a uniform rate of speed. After removal from the chamber and without straightening, the entire length of each 12 inch sample shall be visually inspected for compliance with 3.3.5. After completion of this test both samples of style "SJ" shall be subjected as specified in 4.4.03.2.

4.4.06 Continuity. - All conductors comprising completed wire and cable, and wrapped on shipping spool or reels, shall be subjected to this test. A suitable testing device shall be used to determine compliance with 3.3.6.

4.4.07 Resistance. - All conductors comprising completed wire and cable, and wrapped on shipping spools or reels shall be subjected to this test. A suitable testing device with current limited to 100 ma, max shall be used to determine compliance with 3.3.7.

4.4.08 Solder resistance. - A 12 inch sample shall be cut from each specimen length (see 4.4.09). Each 12 inch sample shall be cut so that the insulation and conductor are flush at both ends. Then remove 1/2 inch of insulation at one end only. At a point one inch from the bared end, each 12 inch specimen shall be bent 90° min around a mandrel of the proper diameter as prescribed in table VIII. The bared end of the conductor then shall be placed into a pot containing 60/40 molten solder so that only 1/8 inch of the end is immersed and held for $5\frac{1}{2} \pm \frac{1}{2}$ second while the temperature is maintained between +590°F and +608°F. Then remove the conductor and visually inspect each 12 inch specimen for compliance with 3.3.8.

4.4.09 Specimen. - From each length of completed wire and cable wound on shipping spools or reels (see 5.1.1) a 5 foot long specimen shall be cut from each end for a total of 2 specimens.

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4.4.10 Delamination. - A 12 inch sample of completed wire or cable shall be cut from each specimen length (see 4.4.9). Both samples shall be cut so that the insulation and conductor are flush at both ends. Both samples shall be placed into an approved air-circulating temperature chamber. The temperature then shall be raised to 482±10°F and maintained for 6 hours min. At the end of 6 hours, both samples shall be removed from the chamber. After allowing to cool, the sample shall be visually inspected to determine compliance with 3.3.9.

5. PREPARATION FOR DELIVERY

5.1 Packaging. - Completed wire and cable, in lengths not less than 110 feet and not more than 1010 feet shall be wound on reels or spools described in 5.1.1 in such a manner that both ends of the length are accessible to permit compliance with 4.4.09. However, deliverable lengths shall be as specified in 3.3.11.

5.1.1 Reels and spools. - Reels and spools shall be of the non-returnable type. Each reel and spool shall have an appropriate diameter for the respective wire size. In no case shall the barrel of the reel or spool have a diameter less than 50 times the nominal diameter of the wire and cable. Reels and spools shall be suitably finished to prevent corrosion under typical storage and handling conditions. The method of attachment of flanges to barrels on metal reels or spools shall be structually equivalent to a full circumferential crimp.

5.2 Packing. - Unless specified otherwise (see 6.1), completed wire and cable shall be delivered in standard commercial containers constructed to insure acceptance by common or other carrier for safe transportation, at the lowest rate, to the point of delivery.

5.3 Marking. - In addition to any special marking required by the contract or order, unit, intermediate packages and shipping containers shall be in accordance with MIL-STD-1169, level "A" - level "C" as specified in the contract or order (see 6.1).

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:
(a) Title, number and date of this specification.
(b) Total footage of each item (see 1.3)
(c) Color requirements for circuit identification (see 3.1.2.1).
(d) Level of marking requirement (see 5.3).

6.2 Definitions.

6.2.1 Objective evidence. - As defined in MIL-STD-109.

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6.2.2 Strand. - One piece of conductive material.

6.2.3 Conductor. - One or more strands combined mechanically in parallel in order that an applied current is divided approximately equally among the strands.

6.2.4 Wire. - One conductor covered by an insulation also known as primary insulation.

6.2.5 Shielded wire. - One wire covered by a conductor.

6.2.6 Shielded and jacketed wire. - A shielded wire covered by an insulation.

6.2.7 Cable. - Two or more wires with optional shield or jacket, combined mechanically and with an optional shield or jacket over the combination.

6.3 Temperature stabilization. - The temperature is considered "stabilized" when three consecutive temperature readings, taken at 5-minute intervals are within $\pm$ 1°F of each other.

6.4 Sampling rate. - The sampling rates herein have been selected to establish a reliability level as required for use of the wire and cable in nuclear weapons.

Custodian:
Army-MU

Preparing activity:
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